

SHAKTI C-Class Dual-Issue Performance Model Experiments

perf-model

August 9, 2026

Executive Summary

The current dual-issue SHAKTI model predicts 10,589,781 cycles and IPC 1.2060 on the generated dual CoreMark trace window. The clean dual RTL ground truth is 10,398,758 cycles and IPC 1.22817; the cycle-stamped run reproduced those app-log counters exactly, so the trace instrumentation is non-perturbing.

Validation gate status: not passed. The dual baseline now has a real per-instruction Δt comparison against RTL, not a proxy. Its accuracy is 93.302% (11,915,999/12,771,407 matching inter-commit deltas), with aggregate cycle error 1.837%. This is below the 97% Phase-1 gate, so the architectural experiments below are useful directional signals but still carry the measured residual timing error.

The backpressure blind spot that motivated this round is fixed for aggregate cycle deltas: the depth-1 s3/s4, s0/s1, s1/s2, and s4/s5 RTL slowdowns are now predicted within a few percent. However, local Δt cadence is still weak for the front-end depth-1 Dhrystone points, so this is not a full mechanism-level closure.

A single-issue floating-point validation addendum now closes the integer-only hole in the validation set. A new dense FP benchmark has 28.575% FLOAT instructions in its measured window. Before FP timing was modelled, the single-issue model reached only 64.612% Δt accuracy on that trace; after adding source-derived FPU latency it reaches 99.863%, with -0.063% aggregate cycle error.

The second memory port alone helps only modestly in this model: +0.698%. Intra-bundle ALU forwarding is the largest individual gain. The best draining two-issue combination found is E+H: intra forwarding + branch, with IPC 1.2953 (+7.406% versus baseline). The quad approximation did not complete/drain in the full-trace sweep and is reported as not completed, not as a measured null. The model-owned branch+branch opportunity count is much smaller than the defective RTL event-52 story suggests.

The IPC 1.7 route analysis gives a sharper answer: IPC 1.7 is not reachable with the current adjacent-pair two-wide structure, because the dependence-only ceiling is IPC 1.6765 on CoreMark. A lookahead window raises the ceiling, but the conservative timed model does not convert that headroom; window 4 regresses to IPC 1.1922 and the best measured lookahead stack reaches only IPC 1.2554.

Experiment Summary

Experiment	Cycles	IPC	Δ IPC	Δ	Verdict
Baseline	10,589,781	1.2060	+0.0000	+0.000%	no effect
A: second memory port	10,516,356	1.2144	+0.0084	+0.698%	helped
B: decouple lockstep	10,517,329	1.2143	+0.0083	+0.689%	helped
C: A+B	10,483,424	1.2182	+0.0122	+1.015%	helped
D: independent retire	10,589,781	1.2060	+0.0000	+0.000%	no effect
E: intra-bundle ALU forwarding	9,968,077	1.2812	+0.0752	+6.237%	helped
F: relax branch next-PC stall	10,589,781	1.2060	+0.0000	+0.000%	no effect
G: symmetric slots	10,589,781	1.2060	+0.0000	+0.000%	no effect
H: second branch unit	10,539,244	1.2118	+0.0058	+0.480%	helped
I: best combination	10,011,897	1.2756	+0.0696	+5.772%	helped
J: quad issue on best	n/a	n/a	n/a	n/a	not completed

Additional Combination Probes

Experiment	Cycles	IPC	Δ IPC	Δ	Verdict
A+E: memory + intra forwarding	10,120,648	1.2619	+0.0559	+4.635%	helped
E+H: intra forwarding + branch	9,859,557	1.2953	+0.0893	+7.406%	helped

Validation

Item	Value
Dual Δt accuracy on generated CoreMark trace	93.302%
Dual Δt matches / compared	11,915,999 / 12,771,407
Dual RTL commit-span cycles from trace	10,398,784
Dual baseline model cycles	10,589,781
Dual clean RTL ground-truth cycles	10,398,758
Dual aggregate cycle error	1.837%
Dual baseline model IPC	1.20601
Dual clean RTL ground-truth IPC	1.22817
Held-out Dhrystone clean dual RTL IPC	1.33 aggregate only

Counter-profile validation. The model emits the RTL-facing counters needed for validation. The local cycle-stamped trace is CoreMark, while the supplied RTL counter profile is from an instrumented Dhrystone run. Absolute counter counts across those benchmarks are not a valid calibration target, so the mem+mem 5% counter gate remains open until either CoreMark RTL counters or a Dhrystone commit trace are produced.

The baseline CoreMark model counter profile is:

Counter	Model value
dual_issued	4,855,836 (45.854% of cycles)
raw_hazard	2,020,130
one_instr	149,723
st3_not_firing	2,022,438
mem_mem_hazard	729,996
mem LL / LS / SS	448,270 / 274,725 / 7,001
branch+branch opportunities	159,885
mispredict	217,257

For reference, the supplied Dhrystone RTL profile was: dual-issued about 50% of cycles, RAW 12,051, one-instruction fetch 4,026, stage3 not firing 4,864, mem+mem 22,510 (LL 6,010, LS 6,000, SS 10,499), and mispredict 2,034.

Held-Out Design-Point Validation

Benchmark	Baseline error	Baseline Δt	Overall floor	Small-effect floor
Dhrystone	-0.427%	99.655%	3.14 pp	0.17 pp
CoreMark	-0.790%	99.150%	1.46 pp	0.64 pp

Dhrystone keeps the original held-out baseline accuracy target. CoreMark baseline Δt is 99.150%, slightly below the requested 99.2% target, while aggregate cycle error remains within 1%. Variant CoreMark Δt is not reported because only the baseline CoreMark commit trace is archived; CoreMark variant rows are aggregate delta checks.

Dhrystone key backpressure points.

Point	Measured	Predicted	Error	Δt
s3s4_1	+91.75%	+93.04%	+1.29 pp	98.40%
s0s1_1	+75.68%	+74.50%	-1.19 pp	67.39%
s1s2_1	+93.55%	+91.55%	-2.00 pp	68.01%
s4s5_1	+90.19%	+90.92%	+0.72 pp	99.98%
s3s4_2	+0.00%	+0.00%	-0.00 pp	99.65%
s0s1_3	+0.01%	+0.00%	-0.01 pp	99.65%
s1s2_3	-0.30%	+0.00%	+0.30 pp	99.34%

	Point	Measured	Predicted	Error	Δt
Coremark key backpressure points.	s3s4_1	+77.62%	+77.97%	+0.35 pp	—
	s0s1_1	+72.86%	+73.21%	+0.36 pp	—
	s1s2_1	+77.29%	+74.63%	-2.66 pp	—
	s4s5_1	+71.48%	+72.23%	+0.74 pp	—
	s3s4_2	+0.00%	+0.06%	+0.06 pp	—
	s0s1_3	+0.88%	-0.02%	-0.91 pp	—
	s1s2_3	-0.11%	+0.00%	+0.12 pp	—

Single-Issue Floating-Point Validation

The prior holdout set was entirely integer. Dhrystone and CoreMark contain no dynamic FLOAT instructions in the measured windows, so the FPU path had not been validated. I added `benchmarks/fpbench/`, a self-contained dense double matrix kernel, instead of importing Whetstone. The reason was practical: `fpbench` uses no libm, has no host-dependent timing harness, and keeps FP work from being constant-folded with volatile matrices and a checksum.

Two independent baseline simulator runs both measured 1,646,171 cycles and 739,818 retired instructions for `ITERATIONS=100`. The measured window’s decoded mix is:

FU class	Instructions	Share
ALU	354,709	47.945%
FLOAT	211,400	28.575%
MEMORY	115,204	15.572%
BRANCH	58,500	7.907%

The FLOAT sub-ops are 211,300 FMA-path operations (`fmadd`, `fadd`, `fsub`, `fmul`) and 100 integer-to-float conversions, so the benchmark exercises the normal FP datapath rather than accidentally compiling to soft-float.

FPU timing. The single-issue makefile defines `spfp` and `dpfp`, but not `hardfloat`; `src/fpu/fpu.bsv` therefore instantiates `fpu_bsvfloat`. The model now detects this from `makefile.inc`. For a `hardfloat` build it reads `fma_stages`, `ftoi_stages`, `itof_stages`, and `ftof_stages` from `src/fpu/fpu.defines`. For the active `bsv_float` build, the FMA path is five cycles: one top-level input FIFO cycle plus the four explicit `fpu_fm_add_sub` state-machine stages before the result is visible. The FPU ready signal lags the result by one cycle because it observes `rg_multicycle_op` and `ff_input.notEmpty`.

Model	Cycles	IPC	Cycle error	Δt accuracy
Before FP latency	812,847	0.9102	-50.622%	64.612%
After FP latency	1,645,141	0.4497	-0.063%	99.863%

The integer baselines do not regress: Dhrystone remains 99.655% Δt accurate with -0.446% cycle error, and CoreMark remains 99.150% Δt accurate with -0.790% cycle error.

FP buffer-depth ladder. The integer-only conclusion that `isb_s3s4` depths 2, 3 and 4 are all free is too broad. Under FP load, depth 2 costs cycles:

Point	RTL delta	Model delta	Error	Δt accuracy
s3s4_1	+42.554%	+45.310%	+2.756 pp	92.118%
s3s4_2	+3.159%	+3.118%	-0.041 pp	99.768%
s3s4_3	+0.000%	+0.000%	+0.000 pp	99.863%
s3s4_4	+0.000%	+0.000%	+0.000 pp	99.863%
s4s5_1	+21.783%	+21.852%	+0.069 pp	99.985%
s4s5_2	+0.000%	+0.000%	+0.000 pp	99.863%

So the area conclusion is revised: `s3s4=2` is not free for FP throughput, while `s3s4=3`, `s3s4=4`, and `s4s5=2` still match the capacity-8 baseline on this workload.

Dual-issue FLOAT pairing. The dual-issue source was checked read-only. `stage2.bsv` allows `ALU+FLOAT`, `FLOAT+ALU`, `CONTROL+FLOAT`, and `FLOAT+CONTROL`, where `CONTROL` is `BRANCH`, `JAL`, or `JALR`. `FLOAT+FLOAT` and unlisted scarce-FU combinations fall through to single issue. The model’s `FLOAT` pairing whitelist matches the RTL source, but there is still no dynamic dual-issue FP validation trace.

IPC 1.7 Route Analysis

Answer. IPC 1.7 is not reachable by polishing the current adjacent-pair two-wide structure. On the dual CoreMark trace, a perfect two-wide adjacent scheduler with infinite functional units, perfect branch prediction, zero memory

latency, no queues, and no structural hazards reaches only IPC 1.6765. That is below the target before any real pipeline costs are reintroduced.

A lookahead issue window is the first mechanism that raises the ceiling: two-wide window 4 reaches IPC 1.9648 and window 8 saturates at IPC 2.0000. To hit IPC 1.7 with window 4, the machine must deliver 86.5% of that new ceiling. The actual dual RTL delivers 73.3% of the adjacent two-wide ceiling and only 62.5% of the window-4 ceiling; the current model baseline delivers 71.9% of the adjacent ceiling. The gap is therefore not only available ILP. It is whether the pipeline can cash that ILP.

Dependence ceilings. The table reports no-same-cycle-forwarding ceilings. The same-cycle-forwarding sanity check reaches the issue width, apart from the final partial issue group when the instruction count is not divisible by the width.

Benchmark	Achieved IPC	2w adj.	2w w4	2w w8	3w adj.	4w adj.
CoreMark dual	1.228	1.677	1.965	2.000	2.070	2.157
Dhrystone dual	1.342	1.876	1.994	2.000	2.552	3.067
fpbench single	0.372	1.981	2.000	2.000	2.578	3.182

The roadmap implication is important: adjacent 4-wide raises the CoreMark dependence ceiling by only 28.6% over adjacent 2-wide, despite doubling issue width. Two-wide window 4 reaches 1.9648, close to the adjacent 4-wide ceiling, so quad issue without an issue/lookahead window is a weak architectural bet.

Full-model route sweep. The dual model baseline error is 1.84%, so any effect smaller than that is below model resolution and is excluded from the recommendation ranking. Lookahead is implemented as a conservative second-slot selector over the next W-1 instructions, with skipped-instruction RAW/WAR/WAW checks and an in-order retire buffer. This is a model experiment, not a validated RTL structure.

Route	IPC	Δ IPC	Ceiling frac.	Candidate checks/cyc	Verdict
Baseline	1.2060	+0.000%	71.9%	0.733	baseline
Intra-bundle ALU forwarding	1.2812	+6.237%	76.4%	0.716	helped; unvalidated
Decouple lockstep readiness	1.2143	+0.689%	72.4%	0.738	below resolution
Independent retire	1.2060	+0.000%	71.9%	0.733	no modeled opportunity
Symmetric slots	1.2060	+0.000%	71.9%	0.733	reversal already handles adjacent pairs
Second memory port	1.2144	+0.698%	72.4%	0.722	below resolution
Relax branch next-PC stall	1.2060	+0.000%	71.9%	0.733	no modeled stall cycles
Second branch unit	1.2118	+0.480%	72.3%	0.734	below resolution
Lookahead window 3	1.2144	+0.695%	63.2%	0.957	below resolution
Lookahead window 4	1.1922	-1.144%	60.7%	1.153	regressed
Lookahead window 6	1.1922	-1.145%	59.6%	1.275	regressed
Lookahead window 8	1.1924	-1.132%	59.6%	1.589	regressed
w4 + intra forwarding	1.2331	+2.246%	62.8%	1.033	helped, worse than intra alone
w4 + intra + branch	1.2502	+3.661%	63.6%	1.016	helped, worse than intra alone
w4 + intra + memory + branch	1.2554	+4.096%	63.9%	0.917	best lookahead stack
w8 + intra + branch	1.2494	+3.598%	62.5%	1.340	no extra win over w4

Two stack configurations were not run on the full trace because they failed to drain on the 200k-instruction prefix: window 4 + decoupled lockstep + intra forwarding + branch, and the full window-8 stack. They are reported as not completed, not as measured null results.

Why lookahead did not convert. The ceiling scheduler says the ILP exists, but the timed model loses it to pipeline coupling. Window 4 reduces modeled RAW rejects from 2,020,130 to 1,489,994, mem+mem rejects from 729,996 to 445,877, and branch-pair opportunities from 159,885 to 97,591. Despite removing those issue rejections, cycles increase from 10,589,781 to 10,712,299. The extra candidates are not free: the model examines 1.153 candidates per model cycle, selects 1,490,752 non-adjacent pairs, then must preserve older skipped instructions with an in-order

retire buffer. The remaining binding constraints are in-order retirement, skipped-instruction ordering, existing queue/backpressure behavior, and the still-large residual RAW count.

The best lookahead stack removes all modeled mem+mem rejects and cuts RAW rejects to 920,951, but still reaches only IPC 1.2554. Plain intra-bundle ALU forwarding without lookahead reaches IPC 1.2812, and the earlier adjacent intra+branch probe reached IPC 1.2953. So the measured route to IPC 1.7 is not “add shallow lookahead to this pipe.” It would require a real decoupled issue queue/rename/retire structure that can extract the lookahead ceiling without turning skipped instructions into retirement and backpressure stalls.

Ranking for RTL effort. Only effects above the 1.84% model-resolution floor are ranked.

1. Intra-bundle ALU forwarding remains the only strong low-to-medium cost candidate: +6.237% alone, +7.406% with the earlier adjacent second-branch probe. It is also the least validated mechanism because `bypass_sources` is not a buildable RTL parameter.
2. A true small issue queue / lookahead scheduler is the only two-wide lever that raises the IPC ceiling above 1.7, but the conservative model-only implementation regresses or underperforms. Treat this as a research direction, not as a ready RTL patch.
3. Do not prioritize a second memory port, lockstep decoupling, or a second branch unit as standalone changes from this evidence. Their individual gains are below the dual model’s resolution floor.
4. Do not pursue quad issue without lookahead. The dependence ceiling says adjacent 4-wide is a poor use of doubled width compared with a two-wide lookahead window.

Calibration Log

Change	Effect	Evidence
Dual trace instrumentation	Added a testbench-only cycle counter to both dual commit-log slots; no core datapath files were changed.	The stamped CoreMark run reproduced the clean dual app-log counters exactly: 10,398,758 cycles and 12,771,408 instructions.
Multi-line <code>app_log</code> parsing	Fixed parser support for SHAKTI logs that print <code>IPC_MEASURE cycles</code> and <code>IPC_MEASURE instret</code> on separate lines, including the space before the colon.	The selected benchmark window now has 12,771,408 instructions, matching CoreMark ground truth.
WAW/WAR pair handling	Kept WAW and WAR co-issue enabled because <code>no_wawstalls</code> compiles out the checks.	Unit tests verify same-rd and write-after-read pairs commit together.
Lockstep scoping	Kept lockstep only for real paired bundles; a single bundle does not wait on slot 1.	Unit tests exercise dependent and independent single/pair cases.
Dual bypass shape	Fixed bypass lookup to inspect slot 1 and slot 0 of both downstream sources, matching <code>bypass.bsv</code> priority <code>src0/slot1 > src0/slot0 > src1/slot1 > src1/slot0</code> .	Full CoreMark dual baseline moved from IPC 1.0357 before this fix to IPC 1.2060.
Memory pairing policy	Split the actual delivered branch (<code>memory_pairing=none</code>) from the proposed second-port experiment (<code>memory_pairing=all</code>).	Run A is now a true second-memory-port experiment instead of the narrower, gated store-involving <code>dual_mem</code> path.
Guarded FIFO backpressure	Modelled <code>mkSizedFIFO</code> as a guarded FIFO where <code>FULL_N</code> is based on cycle-start occupancy, while <code>mkLFIFO</code> remains loopy. Added split compressed fetch-word handling for the single-issue front end.	Dhrystone <code>s3s4_1</code> moved from a near-zero prediction to 322,593 model cycles versus 321,804 RTL; CoreMark <code>s0s1_1</code> is now predicted within 1% aggregate.

Change	Effect	Evidence
CoreMark holdout windowing	Fixed <code>holdout/predict.py</code> so CoreMark variants use the baseline app-log window when only the baseline CoreMark trace is archived.	Removed false instruction-stream drift warnings and replaced invalid variant Δt values with aggregate-only CoreMark validation.
Floating-point timing	Added FP operation classes to the decoder and FPU latency parameters to <code>Model</code> . The active single-issue build uses <code>bsv_float</code> , not <code>hardfloat</code> , so FMA latency is source-derived from the <code>fpu_fm_add_sub</code> state machine rather than from a fitted constant.	<code>fpbench</code> accuracy moved from 64.612% to 99.863% Δt , and the integer baselines did not move.

Analysis

Run A, the second memory port, improves IPC by +0.698%. It removes the modeled non-RAW mem+mem reject counter, but the speedup is much smaller than a first-order lost-slot estimate. The measured interaction is sub-additive, not lockstep-gated: A+B is smaller than A plus B, so those changes remove overlapping bubbles rather than unlocking one another.

Run B improves IPC by +0.689%, confirming that stage3/stage4 coupling is a real limiter. Run C is sub-additive relative to A+B, so the second memory port and decoupling overlap in the bubbles they remove.

Run E is the largest individual improvement at +6.237%. It directly attacks the RAW rejection class and roughly halves the modeled RAW reject count. This is the strongest candidate for RTL work if its implementation cost is manageable.

Run H gives only +0.480%. The model’s independent branch+branch opportunity count is 159,885 on CoreMark, so branch+branch is not the dominant issue in this trace. This is also a useful independent replacement for the defective RTL event 52.

The automatic stacked run I reached IPC 1.2756. The extra probes show why it is not the actual best: A+E reached IPC 1.2619, while E+H reached IPC 1.2953. So the second memory port interferes with the intra-forwarding win in this trace, whereas branch+branch is mildly complementary with intra-forwarding.

Sensitivity

Config	Cycles	IPC	Δ IPC
<code>instr_queue=2</code>	14,681,039	0.8699	-27.868%
<code>instr_queue=4</code>	10,572,287	1.2080	+0.165%
<code>instr_queue=6</code>	10,589,781	1.2060	+0.000%
<code>instr_queue=8</code>	10,590,228	1.2060	-0.004%
<code>instr_queue=12</code>	10,590,778	1.2059	-0.009%
<code>mispredict_penalty=0</code>	10,372,390	1.2313	+2.096%
<code>mispredict_penalty=1</code>	10,589,781	1.2060	+0.000%
<code>mispredict_penalty=2</code>	10,807,038	1.1818	-2.010%
<code>mispredict_penalty=3</code>	11,024,295	1.1585	-3.941%
<code>memory_latency=0</code>	10,589,781	1.2060	+0.000%
<code>memory_latency=1</code>	12,222,907	1.0449	-13.361%
<code>memory_latency=2</code>	14,185,349	0.9003	-25.347%
<code>memory_latency=3</code>	16,155,195	0.7905	-34.450%

What Did Not Work

- Independent retire alone had no effect because actual lockstep stage4 already delivers paired slots to stage5 together.
- Branch next-PC relaxation had no modeled opportunity on this trace: `branch_next_pc_stall_cycles` was zero in the baseline. Treat this as a trace/model result, not proof that the RTL condition is impossible.

- Symmetric slots had no effect for adjacent two-wide CoreMark because the existing reversal logic already moves legal scarce-FU pairs into slot 0.
- Combining decoupled lockstep with intra-bundle forwarding did not drain on a 100k-instruction diagnostic run. The final best combination therefore excludes B when E is selected; this is reported as a model limitation/negative interaction, not hidden.
- Quad issue did not complete/drain in the earlier full-trace timing sweep. The independent dependence-ceiling analysis still shows that adjacent 4-wide is much less attractive than two-wide lookahead.

Experiment Validation Coverage

Experiment	Predicted gain	Validation status	Notes
Baseline	+0.000%	model-only	No specific validation mapping recorded.
A: second memory port	+0.698%	below/near floor	No direct dual memory-port RTL point; fixed memory-latency sensitivity makes the knob active.
B: decouple lockstep	+0.689%	below/near floor	Coupling logic is model-only; effect is near the CoreMark small-effect floor.
C: A+B	+1.015%	near floor	Sub-additive; above small-effect floors but below overall mechanism floor.
D: independent retire	+0.000%	below resolution	Implemented flag has exact zero effect because upstream delivery remains paired.
E: intra-bundle ALU forwarding	+6.237%	no ground truth	Bypass/forwarding has no buildable single-issue design point.
F: relax branch next-PC stall	+0.000%	no modeled opportunity	Baseline branch-next-PC stall counter is zero on this trace.
G: symmetric slots	+0.000%	below resolution	Adjacent two-wide reversal already removes the modeled slot-placement cost.
H: second branch unit	+0.480%	near/below floor	Small model-owned branch-pair opportunity result.
I: best combination	+5.772%	partial	Two-issue stack; inherits E's no-ground-truth forwarding limitation.
J: quad issue on best	n/a	not completed	Full-trace quad approximation did not drain/complete.
A+E: memory + intra forwarding	+4.635%	partial	Shows memory port interferes with E.
E+H: intra forwarding + branch	+7.406%	partial	Best draining two-issue model result; inherits E limitation.

Small-effect floors are 0.17 pp on Dhrystone and 0.64 pp on CoreMark for the single-issue held-out ladder. The dual-issue CoreMark baseline error is 1.84%, so dual design deltas below that threshold are below current model resolution.

Recommendations

1. Prototype intra-bundle ALU forwarding first. Estimated standalone gain: +6.237%; rough RTL cost: medium (operand muxing, bypass select, verification of RAW edge cases).
2. Combine intra-bundle ALU forwarding with a second branch unit only if the branch hardware is cheap. Best measured combination: E+H at IPC 1.2953.

3. Treat the second memory port as a secondary change, not the headline by itself. Estimated standalone gain: +0.698%; rough RTL cost: high (D-cache/store-buffer/arbiter integration). It regresses the E combination, so do not prioritize it before validating with a dual trace.
4. Investigate lockstep decoupling separately from intra-forwarding. Estimated standalone gain: +0.689%; rough RTL cost: medium-high because it touches stage3/stage4 valid/ready and commit bookkeeping.
5. Deprioritize branch+branch hardware until a corrected RTL counter or dual trace says otherwise. Estimated gain: +0.480%; rough RTL cost: medium.
6. Do not commit to quad issue from this evidence. The current approximation did not drain on the full trace and is not a validated 4-wide front end.

Limitations

The model is a timing model over an already committed dynamic trace. It does not compute data values. The validation set now covers small-footprint, cache-resident integer code plus one dense floating-point arithmetic kernel. It still does not model data-cache misses, cache conflicts, instruction-cache misses, wrong-path fetch bandwidth, FP divide/sqrt data-dependent special cases, or CSR/trap behavior that depends on values. The dual CoreMark trace now provides per-instruction RTL commit cycles, but the baseline model only reaches the accuracy reported above; the residual mismatches are concentrated around multiply-heavy and control-flow neighborhoods and should be diagnosed before treating the experiment deltas as RTL-commitment evidence. A Dhrystone commit trace is still missing for held-out dual-issue Δt validation, and no dynamic dual-issue FP trace exists yet.